BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: B.TECH
BRANCH: CSE

SEMESTER : VI/ADD
SESSION : SP/2026

SUBJECT: CS335 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

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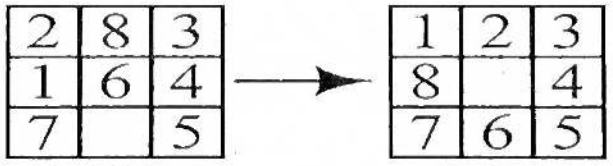
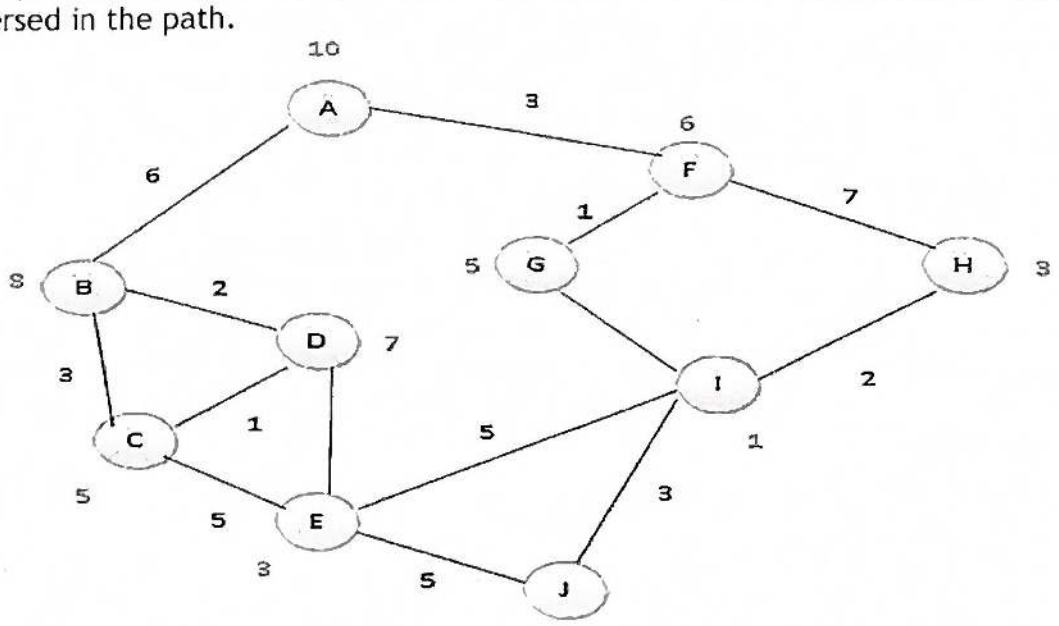
			CO	BL
Q.1(a)	Differentiate between <u>(Any Four):</u> (a) Week AI vs Strong AI (b) Classical vs Modern AI (c) Conventional vs Heuristic Search (d) Propositional logic vs Predicate logic (e) Turing Test vs The Chinese room argument	[0.5x4=2]	1	2
Q.1(b)	What is "**Missionaries and Cannibals**" problem? Give the **production rule** for its solution.	[3]	1	2
Q.2(a)	Solve "**8-puzzle problem**" using Greedy **Best Search algorithm.**	[2]	2	3



Initial State Final Sate

			CO	BL
Q.2(b)	Using **A* algorithm**, find the most cost-effective path to reach from start state "A" to final state "J" for the graph below. The value along with edges represents the actual distance between two nodes and the numbers written on nodes represent the heuristic value (Heurists tic value of End Node, J=0). Be sure to provide calculations for all nodes traversed in the path.	[3]	2	3



PTO

Q.3(a) In CSP problem, solve the following **Cryptarithmatic problem:** [2] 2 4

```
  BASE
 +BALL
 -----
 GAMES
```

In the above equation, the words namely "BASE" and "BALL" are added to produce "GAMES". The algorithm will associate each letter of the given words with a unique number from 0 to 9. For the above input, the output should be -

Letter							
Digits (0-9)							

Q.3(b) Consider any **maximization / minimization** problem, solve this problem through **Hill Climbing method**. Also explain how **Simulated Annealing** can overcome the limitation when **Hill Climbing** gets a stuck in local optimum for this problem. [3] 2 4

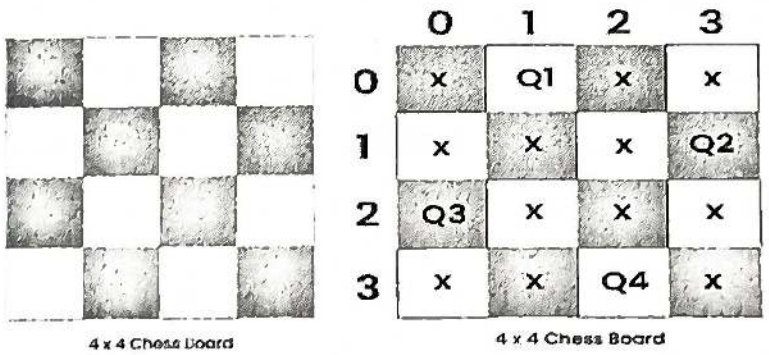
Q.4(a) Why is knowledge representation important in AI? What are some common forms of knowledge representation? [2] 3 2

Q.4(b) Consider the following axioms. [3] 3 5

1. Every bird sleeps in some tree.
2. Every loon is a bird, and every loon is aquatic.
3. Every tree in which any aquatic bird sleeps is beside some lake.
4. Anything that sleeps in anything that is beside any lake eats fish.
5. (Conclusion) Every loon eats fish.

(I) Represent these axioms in predicate calculus.
(II) Convert each formula to clause form.
(III) Prove the conclusion by resolution.

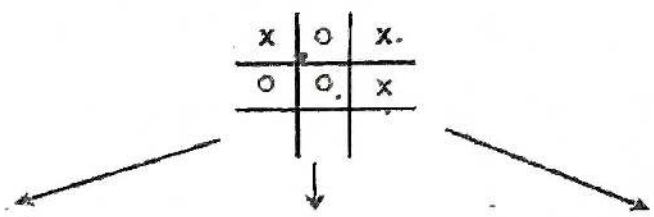
Q.5(a) Place each queen one by one in different rows, starting from the topmost row. While placing a queen in a row, check for clashes with already placed queens. For any column, if there is no clash then mark this row and column as part of the solution by placing the queen. In case, if no safe cell found due to clashes, then **use DFS and backtracking** (i.e., undo the placement of recent queen) and return false. Solve the following illustration of 4 Queens Solution. [2] 4 6



Initial state **Final state**

Q.5(b) A game tree for the **game of tic-tac-toe.** The top node is the initial state, and MAX moves first, placing an X in an empty square. We show part of the tree, giving alternating moves by MIN (O) and MAX (X), until we eventually reach terminal states, which can be assigned utilities according to the rules of the game. How **minimax algorithm solve the tic-tac-toe problem by calculating utility at terminal state and backed up utility at root state for winning X's or O's.** [3] 4 6

Example:



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